

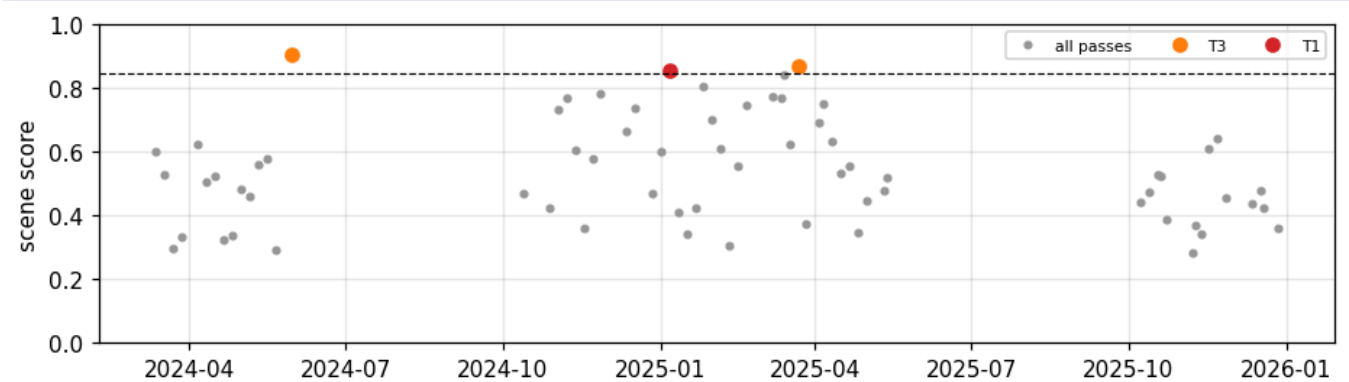
VayuNetra site dossier — Deonar, Mumbai

19.0717, 72.9278 · scan 2024-01-01 → 2025-12-31 · generated 2026-10-01

PRIORITY — methane plume evidence

Evidence

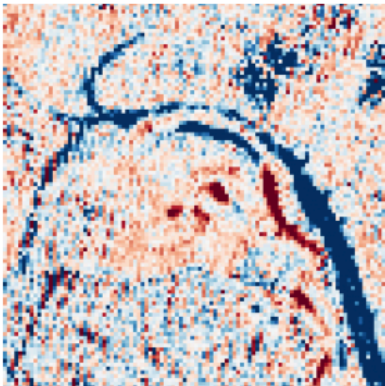
clear passes	flags	T1 methane-confident	T2 probable	T3 surface change	control flags	p vs control	detection limit (t/h)
67		1	0	2	0/65	0.128	nan



Deonar 2025-01-06 (true colour)



methane signal (MBMP)



VayuNetra T1 | 20.1 t/h

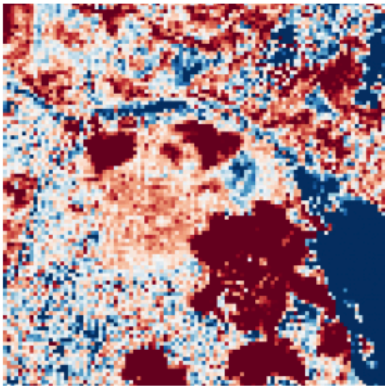


2025-01-06: tier T1 · dB12 -0.052, dB11 +0.003, visible/NIR 0.022 · axis vs wind 31.0°

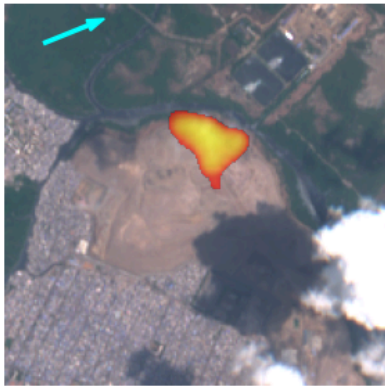
Deonar 2024-05-31 (true colour)



methane signal (MBMP)



VayuNetra T3 | 31.0 t/h

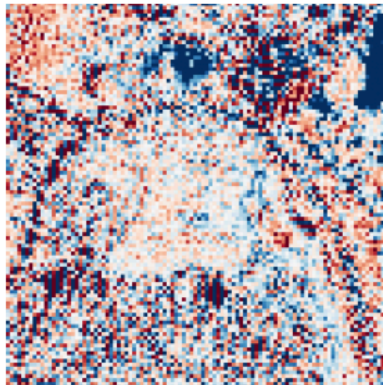


2024-05-31: tier T3 SWIR darkening (moisture / leachate / fresh cover) · dB12 -0.035, dB11 -0.011, visible/NIR 0.133 · axis vs wind 60.0°

Deonar 2025-03-22 (true colour)



methane signal (MBMP)



VayuNetra T3 | rate n/a



2025-03-22: tier T3 SWIR darkening (moisture / leachate / fresh cover) · dB12 -0.013, dB11 +0.055, visible/NIR 0.098 · axis vs wind 69.0°

Emissions

date	tier	q_med	q_lo	q_hi	u10
2025-01-06	T1	20183.0	15054.0	24974.0	5.0

Minimum time-averaged rate: **301 kg/h → 71,249 t CO₂e/yr** (GWP100) · 210,317 t CO₂e/yr (GWP20). Counts only T1 days; all other days assumed zero.

Action plan

1. Confirm within 2–4 weeks: task a hyperspectral satellite (EMIT / EnMAP / PRISMA / Tanager) or fly an OGI-camera drone survey over the flagged area.
2. Install landfill-gas collection (vertical wells + header) on the flagged cell; at 60% capture this avoids ≈41,894 t CO₂e/yr (minimum estimate).
3. Use the gas: ≈0.9 MW electric (7,697 MWh/yr, ≈₹3.8 crore/yr at ₹5.0/kWh); flare whatever the engine can't take.
4. Register the avoided emissions for carbon credits: ≈\$418,945/yr at \$10.0/t.
5. Interim: cover exposed waste and apply a biocover / compost layer on side slopes to oxidise methane; repair cracks and leachate seeps.
6. Satellite sensitivity here: no plume size up to 40 t/h reached 50% detection — screening relies on hyperspectral / ground survey.
7. Long term: continue legacy-waste bio-mining / remediation and divert organics (composting, biomethanation) so less methane is generated.

	value
captured_kgph	180.7
avoided_tco2e_yr	41894.5
power_mwe	0.9
power_mwh_yr	7696.7
power_value_inr_yr	38483293.5
carbon_value_usd_yr	418944.5

Method & assumptions

Sentinel-2 L1C (harmonized) 2 km chips; VayuNetra U-Net (SSL4EO-S12 ResNet-50, fine-tuned on MethaneSET + synthetic plumes), scene threshold 0.84; physics gate (B12-only dimming, wind alignment); IME emission rate with $U_{\text{eff}} = 0.140 \cdot U_{10} + 1.111$; ERA5-Land wind. Assumptions: $\text{gwp}_{100}=27.0$, $\text{gwp}_{20}=79.7$, $\text{capture_eff}=0.6$, $\text{flare_destruction}=0.98$, $\text{ch4_lhv_mj_per_kg}=50.0$, $\text{engine_eff}=0.35$, $\text{power_price_inr_per_kwh}=5.0$, $\text{carbon_price_usd_per_t}=10.0$, $\text{wind_rel_unc}=0.3$, $\text{calib_rel_unc}=0.2$. Satellite estimates are screening-grade and must be confirmed before enforcement or crediting.